

ATLANTIC, IA, USA

WMO: 725453

Lat: 41.407N

Lon: 95.047W

Elev: 360

StdP: 97.07

Time Zone: -6.00 (NAC)

Period: 95-19

WBAN: 14930

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -21.2	(c) -17.8	(d) -26.0	(e) 0.4	(f) -20.8	(g) -22.7	(h) 0.5	(i) -17.3	(j) 14.0	(k) -8.5	(l) 12.3	(m) -5.2	(n) 2.4	(o) 0	(p) 0.549

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.3	33.9	25.1	32.5	24.6	31.2	23.9	26.9	31.9	25.8	30.6	24.8	29.5	5.1	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.8	22.1	31.0	24.1	19.8	29.0	22.9	18.5	27.9	87.3	32.4	82.1	31.0	77.1	29.8	30.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 11.6	(b) 10.3	(c) 8.9	(e) -26.6	(f) 35.9	(g) 2.7	(h) 1.8	(i) -28.5	(j) 37.2	(k) -30.1	(l) 38.3	(m) -31.6	(n) 39.3	(o) -33.5	(p) 40.7		
(a) 11.6	(b) 10.3	(c) 8.9	(e) -26.8	(f) 28.3	(g) 2.4	(h) 1.1	(i) -28.5	(j) 29.1	(k) -29.9	(l) 29.7	(m) -31.2	(n) 30.3	(o) -32.9	(p) 31.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.1	-5.0	-3.1	3.7	10.2	16.5	22.3	24.1	22.2	18.4	11.2	3.4	-3.0		
	DBStd	11.61	6.90	7.09	6.85	5.71	4.88	3.49	3.39	3.19	4.82	5.46	5.62	6.36		
	HDD10.0	1783	465	366	217	66	0	0	0	2	52	207	403			
	HDD18.3	3533	723	599	454	251	93	7	2	6	58	231	448	661		
	CDD10.0	1836	0	1	21	71	208	368	436	379	255	87	9	1		
	CDD18.3	546	0	0	1	6	37	125	180	126	62	8	0	0		
	CDH23.3	5609	0	0	12	86	397	1282	1922	1179	640	88	2	0		
	CDH26.7	2092	0	0	1	21	129	479	803	419	220	20	0	0		
(14) Wind	WSAvg	3.8	4.1	4.1	4.4	4.9	4.3	3.6	2.8	2.5	3.1	3.8	4.0	3.9		
(15) Precipitation	PrecAvg	762	19	24	55	78	97	101	93	95	105	64	36	28		
	PrecMax	1110	55	62	152	196	184	282	217	268	335	123	104	90		
	PrecMin	488	0	2	4	23	40	17	11	8	16	2	0	4		
	PrecStd	152	15	19	40	46	43	70	48	59	76	38	27	21		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.0	17.4	25.0	30.0	33.6	34.7	36.2	35.2	33.8	29.0	22.4	15.9		
		MCWB	7.0	11.2	15.6	18.1	22.2	23.6	27.1	26.2	23.2	19.6	14.8	12.4		
	2%	DB	9.0	12.9	21.3	26.0	30.1	32.7	34.0	32.6	31.4	26.0	17.9	11.3		
		MCWB	5.6	8.5	14.3	16.3	21.2	23.9	26.5	24.9	22.9	18.3	12.8	8.6		
	5%	DB	6.2	9.7	17.6	22.8	27.4	31.2	32.6	30.9	29.1	22.8	15.1	8.0		
		MCWB	3.3	6.8	12.4	14.9	18.9	23.2	25.9	24.4	22.2	16.2	10.5	5.1		
	10%	DB	3.7	6.2	14.8	20.2	25.1	29.8	31.2	28.8	27.3	20.9	12.5	5.3		
		MCWB	1.7	3.7	9.9	13.6	18.2	22.4	25.1	23.3	20.6	15.0	8.6	3.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.0	12.7	17.8	19.7	23.9	26.3	29.1	27.8	25.6	22.6	16.1	13.4		
		MCDB	11.4	16.0	23.0	25.8	30.1	31.5	34.3	32.7	31.1	26.7	20.3	15.2		
	2%	WB	5.7	9.0	15.1	17.9	21.8	25.1	27.8	26.2	24.1	19.5	13.5	8.7		
		MCDB	8.9	13.1	19.0	24.1	28.3	30.5	32.9	31.2	29.0	23.5	17.7	10.4		
	5%	WB	3.7	6.3	12.9	16.0	20.4	24.1	26.6	25.0	23.1	17.6	11.1	5.5		
		MCDB	6.1	9.5	17.5	21.5	25.8	29.5	31.4	29.8	27.7	21.0	14.6	8.0		
	10%	WB	1.9	3.9	10.0	14.5	19.1	23.2	25.5	23.7	21.9	15.6	8.7	3.4		
		MCDB	3.7	6.3	14.7	19.4	24.0	28.4	29.9	27.9	26.3	19.8	12.3	5.6		
(35) Mean Daily Temperature Range	MDBR		9.9	10.2	11.3	12.6	11.6	11.3	11.3	11.5	13.5	12.6	11.4	9.6		
	5% DB	MCDBR	13.3	14.0	16.4	17.2	14.7	12.7	12.3	13.6	14.7	15.6	15.9	13.5		
		MCWBR	10.3	10.1	10.4	9.9	7.5	6.5	7.0	7.8	7.9	9.2	10.7	10.3		
	5% WB	MCDBR	12.6	13.1	14.2	13.7	12.1	11.0	11.0	11.8	12.6	12.7	14.5	12.4		
		MCWBR	9.9	9.9	9.6	8.9	6.8	6.7	7.3	7.6	7.4	8.8	10.6	10.0		
(40) Clear-Sky Solar Irradiance	taub		0.284	0.311	0.326	0.382	0.406	0.413	0.430	0.417	0.380	0.334	0.295	0.284		
	taud		2.424	2.365	2.395	2.258	2.264	2.298	2.239	2.287	2.345	2.462	2.499	2.443		
	Ebn at Noon		875	898	924	887	871	863	843	844	853	862	857	844		
	Edh at Noon		78	97	106	131	134	129	136	126	111	88	72	71		
(44) All-Sky Solar Radiation	RadAvg		1.97	2.95	3.80	4.81	5.73	6.35	6.41	5.54	4.61	3.17	2.18	1.65		
	RadStd		0.17	0.18	0.35	0.35	0.44	0.54	0.45	0.37	0.32	0.38	0.16	0.16		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (53 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page